

Problem ID: gps

You, like so many other people, have probably used the Global Positioning Systems (GPS), but have wondered how it actually works. GPS positioning is based on a set of satellites orbiting earth. All satellites have a synchronized internal clock¹. They continuously transmit the value of their internal clock via radio waves to the surface of earth. A GPS receiver collects simultaneously² measurements from some satellites which are in sight. The time measurements $t_1, \dots, t_m$ collected at time t are typically different, as each satellite has a different distance to the receiver. Since we know the exact orbits of all satellites, we know exactly where each satellite was, when it sent the signal we have received. Based on these positions and the distance, we can compute our position. To compute a unique position, assuming that we are on the surface of the earth, we need at least 3 measurements from satellites.

Your task is, however, not to actually compute your own position, but knowing your position and the current time to determine which signals you received from which satellites. To be precise: You are given your geographical coordinates on earth (which is assumed to be a perfect sphere with a radius of 6371km), a description of all orbits of satellites, and the current time T . You should now determine for every satellite

- whether a signal sent from the satellite at time T will reach you (which is the case if it does not pass through the earth). Remember, radio waves travel in straight lines.
- if the signal does reach you, the time it took for the signal to travel from the satellite to your position. Radio waves travel at the speed of light, which we assume is 299.792.458 meters per second.

We assume that all GPS satellites are orbiting the earth with a fixed speed on perfect circles whose centers are identical to the center of earth.

Input

The input starts with a line containing an integer s ($1 \leq s \leq 10^4$), the number of satellites. Next follows a line containing two real numbers ℓ_o and ℓ_a , the longitude and latitude of your position. ℓ_o is a real number from the range $[-180, 180]$, where positive values represent east and negative values west. ℓ_a is a real number from the range $[-90, 90]$, where positive values represent north and negative values south.

Then follow s lines, where the i th line describes the i th GPS satellite. Each of these lines contains the following real numbers, separated by spaces.

- $\phi \in [-180, 180]$ – the reference longitude at which the orbit intersects with the earth's equator (each orbit intersects the equator (at least) twice). Positive longitude represents east, negative longitude represents west.
- $\psi \in [0, 360]$ – the angle between orbit and the earth's eastbound equator when passing through the equator at longitude ϕ . Values between 0 and 180 indicate northbound movement while values between 180 and 360 indicate southbound movement. If the orbit is coplanar to the equator, the angle ψ will be 0 or 180.
- $r \in [7000, 10^6]$ – the radius of the satellite's orbit in kilometers
- $x \in [0, 1]$ – the percentage of the orbit the satellite has covered at the time when its signal was sent, starting from its intersection with the equator at ϕ . If the orbit is coplanar with the equator, this x will count from longitude 0.

Output

For every satellite output a single line containing `no signal` if a signal sent from that satellite will not reach you. If the satellite's signal will reach you, output the time in milliseconds (rounded to the next full millisecond) that the signal needs to travel from the satellite to your position. The time must be accurate up to a relative or absolute error (whichever is lower) of 10^{-7} .

Sample Input 1

```
3
45.0 45.0
0.0 45.0 7500.0 0.125
0.0 60.0 7000.0 0.3
-50.0 5.0 25000.0 0.3
```

Sample Output 1

```
0.00773245
no signal
0.0690288
```

¹In practice the clocks are not completely synchronous – because of general relativity. In this problem we assume that these effects do not exist.

²Another simplification.

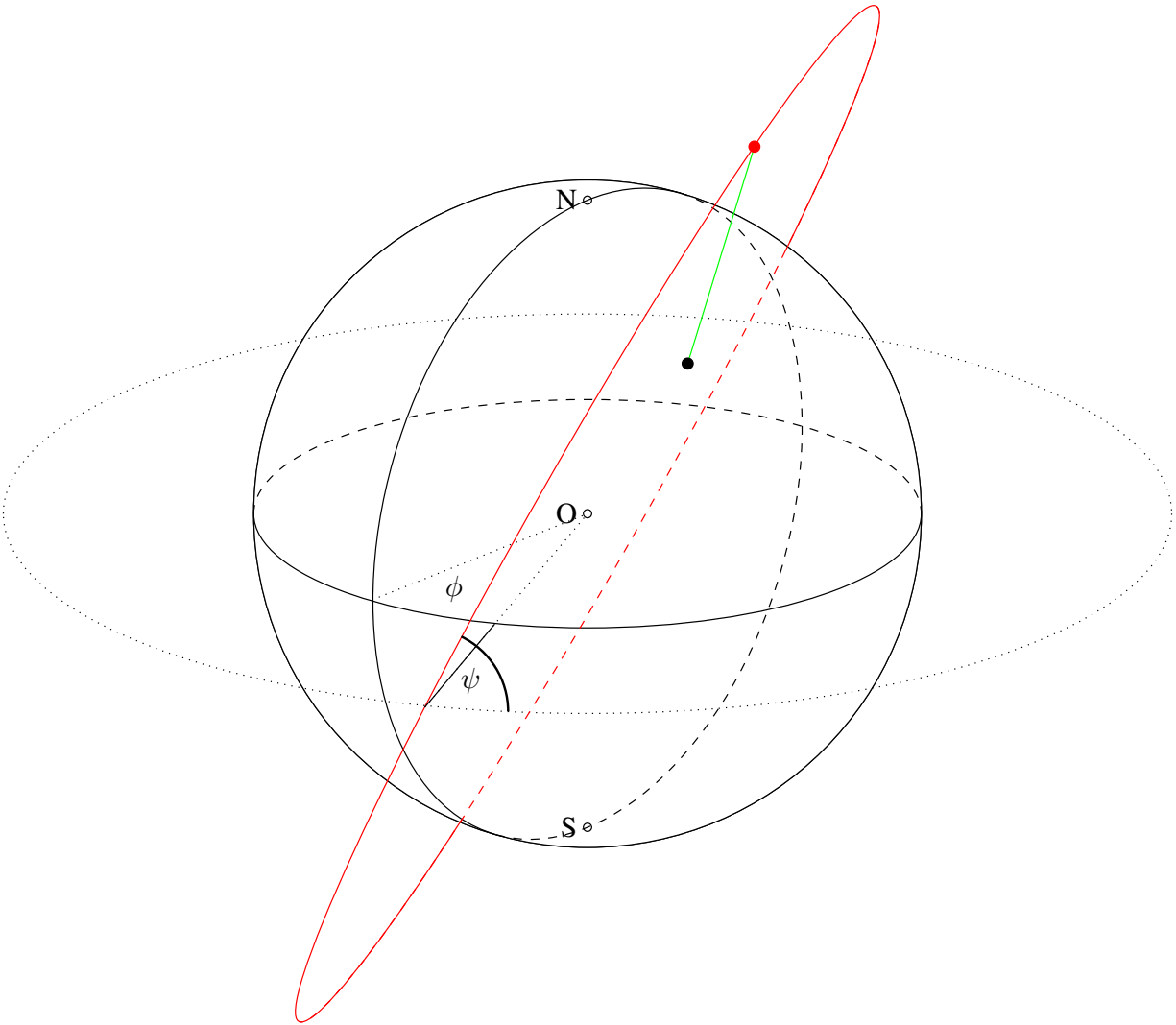


Figure 1: Schematic display of satellite coordinates. The satellites orbit is shown in red, its actual position is the red dot. On earth the equator and the prime meridian are marked by lines. The dotted line is coplanar to the equator and has the same radius as the satellites orbit. The angles ϕ and ψ are described in the sample input. The black dot is your position on earth and you should compute the time necessary for the signal to travel the green line.